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The Potential Competitive Effects of CBDC on Deposits, Payments and Bank Business Models

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I. INTRODUCTION

Money and payments are being disrupted by the digitisation of the economy. New digital assets emerge almost daily and new participants (some organised as firms but frequently as decentralised networks) are entering the process of money creation and intermediation. In this context, central banks are analysing and preparing their policy responses. Perhaps the most important and consequential response would be the potential issuance of a central bank digital currency (CBDC).

A CBDC is likely to have direct competitive effects on several large and important markets, including the market for deposits and the market for means of payment. Understanding these effects is crucial for the policy debates in central banks and government. While competitive effects extend beyond the considerations that central banks could take as motivations to issue (as they rarely have a mandate for overall competition policy), central banks need to understand the potential competitive consequences on these markets as they ponder the trade-offs in the issuance of CBDC. Moreover, in most jurisdictions the CBDC issuance decision will ultimately rest with the government, which, in contrast to central banks, might consider competition as part of their policy motivation. For example, in the United States, the Federal Reserve issues currency, while the Federal Trade Commission is tasked with enforcing antitrust law and achieving competition.

This chapter surveys the nascent literature on CBDC from the perspective of its potential competitive effects. Given the early stages of CBDC development, most of the academic work so far has focused on how different design features influence the potential effects of CBDC using theoretical models or stylised empirical approaches. First, we summarise the main insights of these papers.

This chapter also, however, aims to organise the general lessons of the literature on competition and provide a list of open questions from the perspective of further potential competitive effects of CBDC.

So far, the findings from the literature suggests that the effects of CBDC on commercial bank deposits and lending are not likely to be too large and will be manageable using a variety of policy choices, including the design features of the CBDC product.[1] In this area, most models highlight the importance of the market power that commercial banks hold in the market for deposits, which constrains the supply. Therefore, the reasonable initial intuition that CBDC would disintermediate deposits can be reversed by the fact that CBDC would improve the terms offered by banks on deposits, expanding demand.

Regarding the market for means of payments, the focus of the literature is on the effects of CBDC on established card payment platforms. The effects will be intricate given the network effects of platforms and the complex web of relationships between the parties involved in the card schemes. Our own analysis suggests that CBDC could improve competition in the market for payments and that the optimal pricing of CBDC is likely to consider the existing benefits that established platforms provide to their current users.

Finally, the effects on bank and payment intermediaries' business models are still being analysed by the literature and require further investigation. These effects will depend on how banks and current payment intermediaries get involved in the distribution of CBDC balances and provision of services in the ecosystem.

We divide this chapter by the relevant markets of interest. First, we discuss the effects of CBDC on commercial bank deposits. Then we discuss the market for means of payments and in particular the potential effects on established payment platforms. Finally, we discuss prospective wider effects that have received less attention from research but that will be as important to analyse, in particular the competitive effects of CBDC on firm entry, on bank business models and on new markets.

II. CBDC AND DEPOSITS

CBDC would perform several uses for households. In this section we discuss the impact on the store of value market, with a particular focus on competition with traditional bank deposits. Households lend to banks for safekeeping, usage for payments and for a rate of return. Central banks would likely offer a product that can be both a safe stable store of value and have usage for some payments. They may even pay interest. This section will highlight some of the research on

[1] See, eg, J Li, 'Predicting the Demand for Central Bank Digital Currency: A Structural Analysis with Survey Data' (2021) Working Paper 2021-65, available at: www.bankofcanada.ca/2021/12/staff-working-paper-2021-65/.

the theory of disintermediation of bank lending by CBDC, then some of the nascent empirical work on CBDC adoption, and finally anticipate the effect on competition of CBDC issuance with differing characteristics.

It is common in the industrial organisation literature to assume households value the characteristics and services of the products they consume rather than the products themselves. What characteristics then matter for the household's allocation of store of value when considering a CBDC? First, remuneration, or the direct payments to the holders of CBDC balances, has been discussed extensively in the theory literature. However, paying interest on balances may be unattractive to the central bank as it presents several challenges because of its potential effects on deposits and for the implementation of offline payments. Second, cash, holding deposits, or CBDC allows the household to make payments to retailers online and offline, as well as to peers. A more thorough discussion of the effects on the payments market succeeds this section, but the household derives value in holding balances that can be used for payments. Other characteristics, for which a full discussion is beyond the scope of this chapter, include use in budgeting, privacy from public and private agents, and other services bundled with an account.[2]

A. Market Definition

With the household demand for a CBDC formalised, we turn to defining the relevant market for considering a potential CBDC. Households can hold many products in their portfolio including mutual funds, stocks and bonds. At present, these products are very difficult to use for payments, we therefore exclude them from the relevant market. Next is cash, which in almost all countries is a direct liability of the central bank.[3] For example, an anonymous CBDC that did not pay interest that also credibly protected the privacy of the user up to a certain legal limit would be quite close to cash. An alternative example would be a CBDC that paid interest, required full know-your-customer compliance, and was accessed through the banking system. Such a CBDC would be closer to a bank deposit. For this section, we will restrict ourselves in defining the market as cash and bank deposits that can be easily used in payments.

B. CBDC and Disintermediation of the Banking Sector

The literature on CBDC as a competitor to bank deposits principally began with a discussion of the disintermediation of bank deposits, ie, the substitution

[2] For more information on the definitions and potential effects on the demand for a new CBDC, see ibid.

[3] Privately issued cash like instruments such as pre-paid cards would also be in the relevant market, but we refrain from discussing them.

away from bank deposits to CBDC. This could have the effect of reducing the overall lending in the economy, giving central banks some reason for caution. This comes from the simple accounting identity that loans equal deposits plus equity, and that without an extra source of funding, banks must reduce their lending by the amount of deposits they lose. Keister and Sanches consider the introduction of a CBDC to an economy with perfectly competitive banks (ie, banks without any market power). With competitive banks they show that a CBDC will result in disintermediation. However, they argue that one should weigh the potential efficiency gains in payments from a CBDC against the costs of disintermediation.[4]

Andolfatto adds market power in lending as well as deposit to the discussion, implying a disconnect between the lending rate and the deposit rate.[5] Further banks have access to funding outside the deposit market, namely through the central bank's lending facilities. If the interest rate on CBDC were to induce an increase in the deposit rate, the amount of lending could in fact *increase*. Chiu et al also consider a model with bank market power, except they use a model of monopolistic competition that is more likely to be empirically valid.[6] Indeed, they calibrate the model to US data and find that CBDC could raise bank lending and overall output. These models have, however, taken the demand for consumers as a theoretical object.

We now turn to the new and growing empirical literature on CBDC. Li began considering the choice of consumers to allocate their liquid assets.[7] With household level data she models the introduction of CBDC using the consumers' valuation of its attributes. These include the consumer value of interest income, the ease of use in budgeting, privacy and the bundling of bank services. Inherent in the question is the difficulty of predicting the demand for a product that does not yet exist. With this in mind, Li finds a large range of potential outcomes depending on the design as well as unknown tastes for the product by households.[8]

Whited et al estimate consumer choice from branch level data, then take the demand to a model of banking with richer features than previous papers.[9] In their model, the banks are able to replace a fraction of their deposits with wholesale funding. They predict significant disintermediation especially if the

[4] T Keister and D Sanches, 'Should Central Banks Issue Digital Currency?' (2019) Federal Reserve Bank of Philadelphia Working Paper No 21-37, available at: www.philadelphiafed.org/the-economy/banking-and-financial-markets/should-central-banks-issue-digital-currency-2021.

[5] D Andolfatto, 'Assessing the Impact of Central Bank Digital Currency on Private Banks' (2021) 131 *Economic Journal* 525.

[6] J Chiu et al, 'Bank Market Power and Central Bank Digital Currency: Theory and Quantitative Assessment' (2019) SSRN, available at: ssrn.com/abstract=3331135.

[7] See Li (n 1).

[8] ibid.

[9] T Whited et al, 'Will Central Bank Digital Currency Disintermediate Banks?' (2022) SSRN, available at: ssrn.com/abstract=4112644.

CBDC were to pay interest. Additionally, as wholesale funding is more interest rate sensitive than deposits, the introduction of a CBDC will likely make banks more fragile.

The account fees that consumers pay for their accounts have not been fully studied in the context of the issuance of CBDC. Gibney et al report the nature of the fees and penalties for retail deposit accounts. Some accounts come with monthly account fees, charges for withdraws or transfers and overdraft penalties.[10] A CBDC with attributes similar to a bank account would compete along these margins, especially if the CBDC lacked such fees. Banks would be able to respond to the attributes of a new competitor, potentially causing them to reduce some of their fees to retain consumers. A full account of the welfare and broader implications of this effect is yet to be done. We discuss some of these effects in section IV.

C. CBDC and Stability Considerations

Without an interoperable digital option for households to withdraw their deposits from a bank, the risk of bank runs could be limited in an increasingly digital economy. A CBDC would likely decrease the frictions of converting bank money to public money, as the current option is the physical withdraw of cash or the run to another form of private money, namely another bank. Some, such as Kumhof and Noone, have argued this would have a deleterious effect on stability.[11]

A different argument claims that as CBDC balances would be outside the banking system, at least a portion of the retail payments system would be cushioned from bank runs. This would allow consumers to continue making and receiving payments even during financial stress. Much earlier, Tobin argued that the separation between the functions of deposit and lending would contribute to the effective monitoring of banks' lending through depositors.[12]

Additionally, many papers have discussed the advantages of a CBDC over a bank regulator.[13] The information from a sudden inflow to CBDC would allow the central bank to identify and respond to a run with other policies. A limit on inflow to CBDC or change in the remuneration of CBDC could serve as a tool to stifle such a run directly. Ahnert et al adds a CBDC to the global games literature on bank runs and finds that a CBDC could cause the banks to endogenously

[10] C Gibney et al, 'Banking Fees in Canada: Patterns and Trends', Financial Consumer Agency of Canada (2014).

[11] M Kumhof and C Noone, 'Central Bank Digital Currencies – Design Principles for Financial Stability' (2021) 71 *Economic Analysis and Policy* 553.

[12] T Tobin, 'Financial Innovation and Deregulation in Perspective' (1985) 3 *Bank of Japan Monetary and Economic Studies* 19.

[13] See Keister and Sanches (n 4); S Priazhkina, 'Bank Runs and Central Bank Digital Currency' (2022) Working Paper on file with authors.

change their deposit rates to prevent runs, which might on net reduce the run probability.[14] There are several design choices that could affect the ability of consumers to run from banks to CBDC. For example, a limit on the amount of withdraws could forestall runs. However, this would come at the expense of usage and confidence in CBDC.

III. CBDC AND PAYMENTS

If issued, CBDC may not only serve as a store of value but also as a means of payment. In this section, we discuss the effects that a potential CBDC may have on the market for payments. More specifically, we will focus on how CBDC might affect the competition between non-bank payments service providers (PSPs), such as payment card networks and other fintechs that are entering the payments market.

First, we describe the payment industry together with its main issues and regulatory interventions. Next, we provide an overview of research in the area of platform competition, focusing specifically on payment platforms. We focus on the economic literature that studies the linkage between competition and efficiency in two-sided markets. We show that the literature has found mixed results towards competitive efficiency in two-sided markets, and that procompetitive policies alone might not be sufficient to improve market efficiency. Finally, we discuss the introduction of CBDC as a payment platform and describe the intuition behind recent research and policy work that focuses on the effects of CBDC in the payments landscape.

A. Background and Issues in the Payments Market

The payment industry is characterised by three main features. First, it is two-sided, meaning that there is interdependence between two types of end-users, consumers and merchants.[15] Specifically, consumers choosing a payment method are often concerned about the number of merchants who accept that method of payment. And similarly, merchants' decisions on whether to accept a payment method also depends on the number of consumers choosing that payment method. Examples of common payment methods include cash, debit and credit cards. The existence of this network effect adds complexity to the analysis of competitive efficiency in the payments market, and the effects of a CBDC.

[14] T Ahnert et al, 'Central Bank Digital Currency and Financial Fragility' (2022) Working Paper, available at: www.econ.queensu.ca/sites/econ.queensu.ca/files/CBDCFF.pdf.

[15] See, eg, JC Rochet and J Tirole, 'Two-Sided Markets: A Progress Report' (2006) 37 *Rand Journal of Economics* 645; M Armstrong, 'Competition in Two-Sided Markets' (2006) 37 *Rand Journal of Economics* 668.

Additionally, while there are many new entrants, the industry is relatively concentrated with a few main payment networks, including Visa, MasterCard, Discover and American Express, dominating the market. Further, all of these networks still depend on commercial banks for payment processing and clearing. In addition, some of the networks do not interact with end-users directly but depend on commercial banks for issuing and acquiring services as well. In the so-called 'open system', networks depend on issuing banks for issuing payment cards to consumers and on acquiring banks for merchant-related services that enable merchants to accept payment cards. In these systems, the networks set the interchange fee, which is paid by the acquiring bank to the issuing bank every time a transaction is made.[16]

The payment card industry has been under scrutiny for decades. Regulators and policymakers have noted that consumers are highly subsidised by the card networks, which leads to them using their cards excessively. Merchants and their banks, on the other hand, face high fees that are then passed on to consumers in the form of increased prices. Moreover, these payment networks impose restrictive, and potentially regressive rules, on the merchant side. For instance, quite often merchants must comply with no surcharge rules that forbid them from charging higher prices to consumers based on the means of payment used.[17] This implies that cards that are more expensive for merchants to accept, such as credit cards, will be cross-subsidised by cheaper means of payments such as debit and cash. As high-income consumers are the ones most likely to hold and use cards with higher reward levels that are more expensive for merchants to accept, the cross-subsidies between the payment methods are regressive transfers from low-income consumers to high-income consumers.[18]

Competition authorities and regulators in many jurisdictions, including the United States, Canada, the European Union and Australia, have taken legal action with the aim of improving some of these issues in the payments industry.[19] The most common regulatory interventions, motivated by merchants' complaints, have been in relation to interchange fees. Although following similar objectives of lowering interchange fees, jurisdictions followed different legal and theoretical approaches. Some of the interventions were initiated and executed based on

[16] See M Rysman and J Wright, 'The Economics of Payment Cards' (2014) 13 *Review of Network Economics* 303 for an overview of payment platforms.

[17] See, eg, F Hayashi, 'Discounts and Surcharges: Implications for Consumer Payment Choice' (2012) Federal Reserve Bank of Kansas City Payment System Research Briefing, available at: www.kansascityfed.org/documents/693/briefings-psr-briefingjune2012.pdf.

[18] See, eg, M Felt et al, 'Distributional Effects of Payment Card Pricing and Merchant Cost Passthrough in Canada and the United States' (2021) Bank of Canada Staff Working Paper No 2021-8, available at: www.bankofcanada.ca/wp-content/uploads/2021/02/swp2021-8.pdf.

[19] For a recent summary of regulatory interventions and investigations in different jurisdictions, see F Hayashi and J Maniff, 'Public Authority Involvement in Payment Card Markets: Various Countries, August 2020 Update' (2020) Federal Reserve Bank of Kansas, available at: www.kansascityfed.org/documents/6660/PublicAuthorityInvolvementPaymentCardMarkets_VariousCountries_August2020Update.pdf.

partial analysis while ignoring the two-sided nature of the industry. As a result, the impact of many interventions has sometimes backfired and been accompanied by many unintended effects. Most commonly, they have caused harm to consumers by inverting the traditional business model from a 'merchant-pays' to 'consumer-pays'. Such interventions usually result in reduced revenues for issuing banks, which then react by either increasing existing fees to consumers, such as higher bank account fees, fewer free checking accounts and lower consumer rewards or by introducing new fees to consumers.[20]

For instance, the Reserve Bank of Australia (RBA) aimed to lower interchange fees as it believed that credit card usage was excessively high. However, evidence from Australia shows no substantial changes in card transactions following the intervention in reducing interchange fees.[21] Furthermore, the no-surcharge rule was deemed anticompetitive in Australia. One result, however, was excessive surcharging by merchants to card users, which the RBA subsequently regulated. Another example is the Durbin Amendment in the United States that focused on debit card transactions exclusively by capping only debit interchange fees.[22] The aim of this intervention was to lower consumers' and merchants' costs. However, the Amendment ended up benefiting only some large merchants and it harmed consumers as merchants did not pass through any of their fee savings to them.[23]

B. Profit-Maximising Payment Platforms

Although the payments market is currently concentrated, with only a few dominating platforms, there are several smaller players and potential entrants that bring competition into the market. The objective of this section is to use the economic literature to answer one main question: can competition in the payments market lead to market efficiency (ie, maximising total welfare)? The key concern behind this question is whether platforms would fully internalise the network externality (an increase in value to other users when a user joins the platform): in essence, how a user's utility or a merchant's profit depends on the composition of a network's users.

[20] See, eg, C Howard et al, 'The Effect of Regulatory Intervention in Two Sided Markets: An Assessment of Interchange-Fee Capping in Australia' (2005) 4 *Review of Network Economics* 350.

[21] See especially, R Hayes, 'Is Price Regulation of Payment Card Associations Effective? Evidence from a Dramatic Policy Experiment' (2010) SSRN, available at: papers.ssrn.com/sol3/papers.cfm?abstract_id=1546869.

[22] For details, see Press Release, 'Federal Reserve issues a final rule establishing standards for debit card interchange fees and prohibiting network exclusivity arrangements and routing restrictions' Board of Governors of the Federal Reserve (29 June 2011), available at: www.federalreserve.gov/newsevents/pressreleases/bcreg20110629a.htm.

[23] See, eg, B Hubbard, 'The Durbin Amendment, Two-Sided Markets, and Wealth Transfers: An Examination of Unintended Consequences Three Years Later' (2013) SSRN, available at: papers.ssrn.com/sol3/papers.cfm?abstract_id=2285105.

We start with the most optimistic answer to the question. Jain and Townsend show that competition among platforms forces them to internalise the aforementioned externality, and consequently, leads to Pareto efficiency.[24] To prove this, they consider a general equilibrium model with an intermediary that creates an infinite number of potential platforms. Each platform specifies the number of merchants and consumers that it anticipates accommodating.[25] The intuition behind the result that the competitive equilibrium is efficient is simple. Assuming the intermediary can create platforms with all possible composition of buyers and sellers, they will expand the commodity space to incorporate the network externality in a manner suggested by Arrow.[26] The user prices for joining platforms, which each user takes as given, fully internalise the users' marginal utility gains from altering the size of the platform.

However, Jain and Townsend rely on the crucial assumption that the market is perfectly competitive, which might not be a reasonable assumption in the payments market.[27] They also study the case under which the intermediary is a monopoly. As expected, in the monopoly equilibrium the intermediary will use its market power to charge higher prices, leading to an inefficiency.

A potentially more realistic model on the payments market considers an oligopolistic economy where each platform has market power and sets prices to maximise its own profit. Armstrong considers such a market with two platforms, which sell horizontally differentiated consumption goods delivering different intrinsic values as well as two-sided network benefits to consumers.[28] Armstrong characterises the equilibrium in which both platforms charge a flat price.[29] A key insight from Armstrong's analysis is that the network effects make the market more competitive compared with a market without network effects. This is because the network effect will generate a negative feedback loop. For example, when a platform raises its price on the consumer side, consumers will leave that platform and join the rival platform, which further drives merchants of that platform to leave and join the rival platform even though the merchant side prices have not changed.

Based on the insight from Armstrong, network effects are procompetitive – a larger network effect will lead to a more competitive market – raising the question of whether it will lead to a more efficient market. The answer is: not always. The reason is twofold. First, even with a considerably large network effect, prices in the oligopolistic equilibrium are still distorted by the market power.

[24] A Jain and R Townsend, 'The Economics of Platforms in a Walrasian Framework' (2021) 71 *Economic Theory* 877.

[25] ibid.

[26] KJ Arrow, 'The Organization of Economic Activity: Issues Pertinent to the Choice of Market Versus Nonmarket Allocation' (1969) 1 *Analysis and Evaluation of Public Expenditure: The PPB System* 59.

[27] See Jain and Townsend (n 24).

[28] Armstrong (n 15).

[29] ibid.

Second, in order to focus on market-sharing equilibria,[30] Armstrong assumes that network effects are small compared with the degree of differentiation between the two platforms, and that the differentiation affects the platforms' market power in a positive way. This means that the more differentiated these two platforms are, the more market power they have to set higher prices.[31] Therefore, while the network effect is procompetitive and drives platforms to set lower prices, the greater differentiation between platforms will distort prices in the opposite direction. On the other hand, if platforms are allowed to have a large network effect without restrictions, competition between platforms will have the tendency towards a monopolistic market. In this case each consumer and merchant in the market chooses to join only one platform, which leaves the other platform with no incentive to stay in the market, and market power still distorts prices.

Surprisingly, it is possible that the procompetitive tendency of the network effect can be reversed if a pricing formula other than flat pricing is used. This is mentioned in Armstrong in the discussion of two-part tariffs and further studied by White and Weyl by focusing on a special case of this type of tariff.[32] Armstrong extends its model to accommodate more flexible pricing, a two-part tariff, under which users pay a fixed fee together with a marginal price for each user on the other side who joins the platform.[33] The equilibrium analysis shows that marginal prices enable platforms to ease the competition so that they can charge higher prices as the network benefits become larger.

White and Weyl study a special case in which users pay a marginal price exactly equal to the network benefit they receive from each user participating on the other side.[34] They then propose a solution concept, Insulated Equilibrium, based on the idea that users from each side facing this special two-part tariff will have a dominant choice of platform participation. In their model, each user's decision on which platform to join is independent of the participation decisions on the merchant's side. Following the solution concept, platforms in a duopoly commit to using this sophisticated pricing strategy and, consequently, they are able to shut down the negative feedback loop mentioned in Armstrong and reduce the competitive effects of network benefits.

While the theories mentioned above apply generally to any two-sided market, there is also a growing literature studying this topic in the context of payments market.[35] Many papers in this literature have focused on the high interchange fees. Guthrie and Wright are among the first in the literature to study whether

[30] A market-sharing equilibrium means platforms share the market. The opposite is the tipping equilibrium, in which one platform dominates the market.

[31] See Armstrong (n 15).

[32] EG Weyl and A White 'Insulated Platform Competition' (2016) SSRN, available at: papers.ssrn.com/sol3/papers.cfm?abstract_id=1694317.

[33] See Armstrong (n 15).

[34] See Weyl and White (n 32).

[35] See Rysman and Wright (n 16) for a review of this literature.

competition between payment platforms can lower interchange fees.[36] However, their analysis shows that the results are case dependent and that, in some cases, increased competition might even lead to a higher interchange fee: in a more competitive environment, payment platforms may pay more rewards to consumers while increasing the merchant fees. Chakravorti and Roson follow a different modelling approach and find that competition in the payments market unambiguously reduces equilibrium prices.[37] However, they also admit that competition may not always lead to welfare improvements because it might lead to more distortions in the interchange fees. For example, credit card companies may pay more cash-back benefits to consumers while increasing the merchant fees.

C. CBDC as a Public Payment Platform

There are abundant examples across many markets where public and private products coexist. Those examples include, more specifically, goods provided in platform-like set-ups like schools and hospitals. Yet, if issued, a CBDC would be a government-run product competing directly with existing payment platforms. While cash competes as a means of payment, it does not have the ability to cross-subsidise the way electronic payments do and cannot be used online. In this way, a CBDC would open a new competition front between public and private means of payments. In this section, we focus on discussing various potential competitive effects of CBDC on the payments industry.

i. CBDC and Payments at the Point of Sale (PoS)

In this use case, the ecosystem of firms includes the card networks (which establish the connections between consumers' and merchants' banks), the commercial banks (which issue the deposits used to transfer value between the parties and coordinate with the networks to issue the payment cards to depositors), and merchant acquirers (who provide payment services to merchants, like terminals). There are a multitude of prices and fees that could be affected by the introduction of a CBDC. But perhaps one of the most important price channels through which CBDC could affect the established payment platforms is via its effect on the interchange fee charged today by debit and credit card networks. A CBDC could enter the market as a competing platform by setting its own fees and rules. The incumbent card networks might respond by lowering their fees or modifying

[36] G Guthrie and J Wright, 'Competing Payment Schemes' (2007) *55 Journal of Industrial Economics* 37.

[37] S Chakravorti and R Roson, 'Platform Competition in Two-Sided Markets: The Case of Payment Networks' (2006) *5 Review of Network Economics* 118. Different from Guthrie and Wright, where it is assumed that platforms are non-profit and provide identical network effects, Chakravorti and Roson focus on profit-maximising platforms and allow them to offer different network benefits to different consumers.

the terms they set on the members of the ecosystem (such as 'honour all cards' and 'no-surcharge' rules). However, the incumbents might further distort their fee structures – giving more rewards to consumers and elevating merchant fees – in response to the entry of a CBDC platform. The equilibrium outcomes are likely to be complex, in particular because the two-sidedness of the market will play a prominent role.

ii. CBDC in Online Markets

The next frontier of competition issues will likely arise in online payments. At present, no outside or public money is offered to consumers for online transactions. Therefore, further to helping moderate interchange fees for PoS transactions, a CBDC as an online means of payment could provide end-users with an alternative to credit cards and other emerging methods of payment. According to Usher et al, CBDC for online transactions could provide the same safety and affordability that cash has offered in the offline world.[38] It is crucial to point out that while CBDC has the potential to be procompetitive in these markets, the actual impact will greatly depend on the response of the existing incumbent payment platforms. If CBDC entry were to trigger a wave of mergers and acquisitions and result in a more concentrated payments industry, then the effects might be weaker. More research is needed to understand which effect has the potential to dominate if such a scenario unfolds.

iii. CBDC as a Benevolent Payment Platform

If the CBDC payment system were to operate in the same way as the current private payment platforms, ie, focusing on maximising profits, it would exert a similar competitive effect as any other private platform and the same results as reviewed in the previous section would hold. CBDC, however, might operate as a benevolent payment platform aiming to maximise welfare instead of profits. Most of the literature so far has explored the effects of competition between profit-maximising platforms. In contrast, Liu et al focus on the payments aspect of CBDC by introducing CBDC as a benevolent payment platform that maximises total surplus instead of profits in an oligopolistic market competing with private payment platforms.[39] They find that the competitive equilibrium with a benevolent payment platform leads to higher social welfare than the equilibrium with profit-maximising platforms only. In addition, the CBDC platform faces a trade-off between attracting users to the CBDC platform and accommodating some users' preferences for using the private payment

[38] A Usher et al, 'A Positive Case for a CBDC' (2021) Bank of Canada Staff Discussion Paper 2021-11, available at: www.bankofcanada.ca/2021/07/staff-discussion-paper-2021-11/.

[39] Y Liu et al, 'CBDC and Payment Platform Competition' (2022) Working Paper on file with authors.

platform (due to heterogeneity in consumer taste). For that reason, a CBDC payment platform should not only consider undercutting the private platform but adjust its price to the level that users might still choose to use the private platform. This suggests that setting the optimal prices of using the CBDC payment platform may not be as straightforward as setting the fee consistent with a cost recovery objective, as is done in other payments systems provided by central banks. This finding also implies that while cash is an obvious alternative payment option, it might not necessarily be welfare improving because the price to pay with cash is zero to both sides, therefore precluding the cross-subsidies allowed by electronic payments.

iv. CBDC and Layers of Intermediation

One of the main features of the non-bank payment service provider (PSP) industry, as previously mentioned, is that it depends on commercial banks to act as intermediaries for completing the settlement part of their payment business. More layers of intermediation usually translate to higher mark-ups for end consumers, as financial participants in each layer need to be compensated for their services and may exert market power. Halaburda et al explore this double marginalisation problem in the card payments market.[40] One way in which CBDC could improve competition in payments is if it were to reduce the layers of intermediation. Andolfatto notes that one clear advantage for CBDC, compared with other PSPs, is that it will have access to wholesale payment rails.[41] The actual procompetitive potential of CBDC will depend on the design choices that central banks will make especially with regard to their CBDC's distribution model. In policy circles, a distinction has grown between CBDCs issued directly to end-users, named a 'unilateral' or 'single-tier' system, and an 'intermediated' or 'two-tier' system where banks and other intermediaries would deal with the end-users.[42] Bossu et al discuss this distinction and list the functions the public or private sector would have to divide in the two-tier case.[43] While a one-tier distribution model is still being considered, most central banks are discussing the implementation of a two-tier or intermediated approach for CBDC. This implies that central banks might still depend on other financial institutions (FIs) to provide end-user services. Potential intermediaries include existing commercial banks or other regulated FIs and fintech companies as well as public entities.

[40] H Halaburda et al, 'Interchange Fee, Market Structure and Excessive Intermediation in the Payments Market' (2022) Working Paper on file with authors.

[41] D Andolfatto, 'On the Necessity and Desirability of a CBDC' (2021) *Mimeo*, available at: gceps.princeton.edu/wp-content/uploads/2021/11/21oct_Andolfatto-paper_CBDC4US.pdf.

[42] China's CBDC pilot is a two-tiered system where the banks offer consumers access to CBDC wallets.

[43] W Bossu et al, 'Behind the Scenes of Central Bank Digital Currency' (2022) IMF FinTech Notes No 2022/004, available at: www.imf.org/en/Publications/fintech-notes/Issues/2022/02/07/Behind-the-Scenes-of-Central-Bank-Digital-Currency-512174.

Nonetheless, CBDC may still have the potential to exert a competitive role in the industry if it entails fewer layers of intermediation, or more efficient intermediation. CBDC might allow access for more efficient intermediaries across the distribution chain compared with other PSPs or in comparison to the current intermediated distribution models that most central banks employ for cash.

v. CBDC versus Other Interventions

Another aspect to consider is how CBDC issuance would differ in impact and interplay from traditional regulatory or antitrust interventions. First, a CBDC would be an alternative payment platform for customers and merchants and not simply a cap or restriction on existing platforms' fees. This could potentially help bring down the fees charged by the established networks as end-users would be granted with an additional payment option. This is crucial, especially if regulation or CBDC entry leads to market consolidation by incumbent payment platforms. In this way at least, the end-users that adopt the CBDC platform could benefit directly. Second, CBDC could be a powerful tool in restricting the scope of some of the unintended effects that have risen from other regulatory interventions. For instance, if caps are put on interchange fees, issuing banks can simply find alternative ways to increase their lost revenues by increasing or introducing other fees. If only private platforms are competing in the market, then doing so would be simple as all platforms have the same incentives. If, however, there were a public platform in the market that did not have such incentives then it might be more difficult for private platforms to do so without losing consumers. Therefore, CBDC could work in concert with regulation to achieve more efficient policy outcomes.

vi. Potential Limitations to CBDC's Potential Competitive Effects

CBDC will not offer rewards and benefits in the way that credit card networks offer their consumers. This might make it difficult for CBDC to attract users or exert competitive pressure especially in the credit card market. Finally, CBDC could also face the same entry barriers as private entrants in the payments market. Existing network effects, where consumers value the methods of payments that are widely accepted by merchants and vice versa, as well as privacy concerns from the public, might limit the adoption and competitive effects of CBDC.

IV. CBDC AND THE BUSINESS MODEL OF BANKS

The competitive effects in the markets discussed earlier will interact with each other. Perhaps the most evident channel of interaction will be through the effects on the business model of banks as they are both issuers of deposits, issuers of cards and participants in the electronic card schemes. In this section, we discuss some of the potential competitive effects on the business models of banks.

The question of effects on business models of banks goes beyond the effects on lending from a substitution away from deposits towards CBDC (discussed in section II). Bank deposits offer services beyond store of value and payments; further, deposits are only a part of a suite of services that banks provide to their customers. The substitution towards CBDC might also affect the product bundle that traditional banks offer. CBDC might also affect banks through its effects on the complementarity between deposits, payments, lending, investments and data. For example, the adoption of CBDC as a means of payment could affect the economies of scope between payments data and consumer and business credit, which has been documented empirically quite extensively.[44]

While the literature on CBDC has not yet explored this channel, some recent work is exploring the effects of fintech competition on banks which provides some guidance as to the potential effects of CBDC. For example, Parlour et al study the impact of fintech competition in payment services when a bank uses payment data to learn about consumers' credit quality.[45] They find that competition from fintech payment providers disrupts the information spillover. In their model, a signal about a consumer's credit quality can be extracted from payment transactions. In this way, a bank has less precise information of a loan applicant's credit quality if the applicant has made payments via a fintech rather than through the lending bank. They show that if the assumption of less precise information is relaxed, bank lending would increase, however the effect on consumer welfare would be ambiguous. If CBDC competes with banks for payments, its informational disruption effects for banks could be similar.

At the outset, the direction and overall magnitude of these effects is unclear because banks could maintain customer relationships (or even offer new services related to the CBDC product) if they become involved in the CBDC ecosystem. For example, banks could be providers of the electronic wallet services that allow customers to hold CBDC balances. In this case, the customer relationship could be largely unaffected even if the balance that a customer uses for payments is no longer issued by the commercial bank. If the complementarity between different bank products and services is determined mostly by the customer holding a relationship instead of the amount of balances she holds, then we could expect the effect of CBDC on banks to be limited via this channel. Theoretical and empirical work will be required to understand and quantify these effects.

V. CONCLUSION

This chapter surveys the emerging literature on CBDC and discusses the potential competitive effects on three areas: the market for deposits; the market for

[44] See, eg, L Mester et al, 'Transactions Accounts and Loan Monitoring' (2007) 20 *Review of Financial Studies* 529.

[45] C Parlour et al, 'When FinTech Competes for Payment Flows' (2020) SSRN, available at: papers.ssrn.com/sol3/papers.cfm?abstract_id=3544981.

means of payments; and on the business models of the established intermediaries in these two markets. The literature so far suggests that the effects on commercial bank deposits and lending are likely to be manageable while the overall effects on payment intermediaries and bank business models still require further investigation.

Looking ahead, however, CBDC is likely to have competitive effects beyond those we considered here. One such effect could be enabling the entry of new types of intermediaries to the payment ecosystem, in particular firms that are not deposit-taking institutions. These new types of firms would not compete directly with banks in issuing deposits but in offering associated financial services. This has the potential to increase the competition for customer relationships.

One last aspect that will be relevant to consider in future research is how the competitive effects discussed might vary as cash demand and usage wanes in the future. Recent literature has highlighted the role that cash has in limiting market power in payments, therefore it will be important to explore if the competitive effects of CBDC discussed above would substantively change in the absence of cash.